Hsuan-Fu (Jeff) Hsu

Embedded Systems and AI Engineering | C/C++, Python, STM32, Raspberry Pi, AWS

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science — Sept 2020 - June 2024

SKILLS

Proficient: C/C++, Python, C#, Git

Familiar: Raspberry Pi 4, Arduino Nano 33 BLE Sense, Arm Cortex-M4, I2C, UART, BLE, Renode, GDB, Linux, OpenGL, GLSL, AWS (Lambda, SQS, DynamoDB, CDK), Docker, pytest, GitHub Actions, SQL, TypeScript, PyTorch

Exposure: TensorFlow, tsfresh

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Co-developed a portable NIR sensing device with two teammates, using an **Arduino Nano 33 BLE Sense** and an AS7263 sensor connected over I2C. The device sampled 810/860 nm signals at 34 Hz and transmitted readings over BLE.
- Wrote the **Python** pipeline that processes NIR signals from 30 participants. Our best model estimated uric acid levels with 10.3% MAPE in leave-one-subject-out testing.
- Received an individual NSTC research grant and placed fourth in NTPU's project competition as part of a three person team.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- As the sole developer, built a birth grant application system in **ASP.NET Core MVC** and **SQL Server** and rolled it out to 29 government offices.
- Tuned SQL Server processing and storage, which cut database latency by at least 50% and resource usage by at least 30%.

PROJECTS

Cortex-M4 Emulator and Bare-Metal Firmware — Nov 2026 - Present
C, Arm Thumb-2, STM32F446, Renode, GDB | GitHub

- A **Cortex-M4** emulator in C that decodes Thumb-2, handles exceptions through NVIC and SysTick, and runs bare-metal STM32F446 firmware with no board. A **GDB** stub supports breakpoints and single stepping.
- Checked it against **Renode** one instruction at a time: 33 of 35 test programs matched, as did the first 16,439 instructions of the firmware.
- The no-HAL firmware uses interrupt driven **UART** with ring buffers and **I2C** drivers for a BME280 and an SSD1306. Unit, emulator, and Renode tests run in GitHub Actions CI, but not yet on a real board.

Classroom AC Control System — USC IEEE HackIoT (team of 4) — Oct 2026
Raspberry Pi 4, Python, YOLO, DHT11, Supabase | GitHub

- Set up a **Raspberry Pi 4** with teammates to count classroom occupants using YOLO at approximately 7 FPS and log DHT11 temperature and humidity readings to Supabase every 30 seconds.

3D Rendering Engine — Oct 2022 - Jan 2023; rebuilt Nov 2026
C++, OpenGL, GLSL, CMake | GitHub

- Rebuilt a **C++**/**OpenGL** renderer with an OBJ/MTL parser and vertex deduplication, reducing a test model's vertex count by 71% and parse time from 108 ms to 56 ms.
- Fuzz tested the OBJ and glTF loaders under ASan/UBSan, including more than 710,000 OBJ inputs, and found no issues. Also added Cook-Torrance PBR, PCF shadows, and glTF 2.0 loading.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, 2026, doi: 10.1109/THMS.2026.3724413.